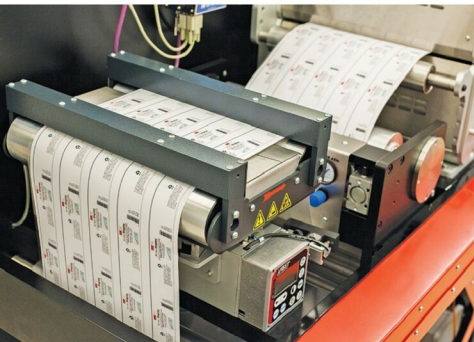




Integrated printed biosensor developed by Acreo (Courtesy: Acreo Swedish ICT)

ics. Such solutions require a range of functionality, from complementary logic, power sources, displays and memory devices to the integration of a wide variety of sensor types. While much of this can be achieved through direct, additive printing of solution based inks, some complex computational functionality such as high-resolution analogue-to-digital conversion and wireless communication are better done with conventional silicon based electronics, today.

While this is bound to change in the future as capabilities of printed electronics improve, today it becomes necessary to combine the benefits of printed, flexible, distributed sensors and the performance of silicon electronics, if we want to achieve the best efficiency in some applications. In order to achieve this, Xerox PARC, an expert in large-scale, flexible sensor systems, has come up with a hybrid fabrication platform.

Presented at Nano Bio Manufacturing Consortium (NBMC) Wearable Electronics Workshop last year, this technique allows for printed electronic devices to be used alongside a minimal set of pre-fabricated microelectronic components. By

describing how to mechanically and electrically interface printed and microelectronic components in a system that wirelessly transmits data from printed light and temperature sensors, researchers showed that it is possible to combine small, low-profile integrated circuits (ICs) into flexible, custom, high-performance sense-and-transmit systems without affecting the mechanical flexibility of the final system.

### Mass manufacture of classic designs

If only it was possible to mass-print device designs, it would mean a great relief for numerous electronics engineers stumped with how to take their printed electronics research out of the lab into large-scale production.

One step towards this was taken by PARC and Clemson University last year in a project facilitated by Flex-Tech Alliance. The project explored how to scale up and print functional devices on a commercial printing press.

PARC contributed brilliant device designs amenable to inkjet printing, which it generally printed using a lab-scale gravure printing press not

suitable for high-speed, roll-to-roll throughput. Clemson University came into the picture with its seven-station Omet printing press, a modular platform. The goal of the project was to achieve multi-layer device printing with a gravure or flexo-hybrid approach.

This involved several customisations to the press, inks and design. For starters, the team engineered a chambered doctor blade system on Omet press. The chamber enables users to seal a small volume of ink, less than 50ml, against the cylinder, unlike normal gravure press configurations that use a large amount of ink to fill the pan. This restricts ink usage, saving money in the bargain.

A variety of inks were tested, and numerous functional multi-layer devices such as capacitors, diodes and thin-film field-effect transistors (FETs) were printed. Device parameters were optimised using high-speed optical metrology, and resistivity of around 4X bulk silver for features approximately 50nm thick at speeds of around 60m/min (200-feet/minute) were achieved, with very high yield.

### Up in 2016: Printed sensors to watch engines and motors

University of California at Berkeley and California Polytechnic State University in San Luis Obispo, both in the USA, are working towards developing fully-integrated, printed, self-rechargeable wireless sensor nodes for engine and motor condition monitoring.

Condition based monitoring (CBM) is in great demand today and is required by most industries. The advantage of using flexible electronics for this purpose is that, devices can be designed to fit unique geometries. In this case, the team plans to use a flexible substrate, so that the device can be rolled up into a cylinder with a small footprint. This will be a big jump from the use of bulky and heavy equipment that cannot be integrated into the limited space around engines and motors. By using